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Studierichting: Informatica
Oefeningenreeks 6B nr. 21.4

Bereken de n-de partieelsom van $\dfrac{2(n+2)}{n(n+1)(n+3)(n+4)}$

$= \dfrac{A}{n} + \dfrac{B}{n+1} + \dfrac{C}{n+3} + \dfrac{D}{n+4}$

$A = \dfrac{2(0+2)}{(0+1)(0+3)(0+4)} = \dfrac{1}{3}$

$B = \dfrac{2(-1+2)}{-1(-1+3)(-1+4)} = -\dfrac{1}{3}$

$C = \dfrac{2(-3+2)}{-3(-3+1)(-3+4)} = -\dfrac{1}{3}$

$D = \dfrac{2(-4+2)}{-4(-4+1)(-4+3)} = \dfrac{1}{3}$

$= \dfrac{1}{3n} - \dfrac{1}{3(n+1)} - \dfrac{1}{3(n+3)} + \dfrac{1}{3(n+4)}$

Als we dit dan invullen voor $n = 4$, dan zien we dat deze rij teleskoperend is.

$n = 1$

$\dfrac{1}{3} - \dfrac{1}{6} - \dfrac{1}{12} + \dfrac{1}{15}$

$n = 2$

$\dfrac{1}{6} - \dfrac{1}{9} - \dfrac{1}{15} + \dfrac{1}{18}$

$n = 3$

$\dfrac{1}{9} - \dfrac{1}{12} - \dfrac{1}{18} + \dfrac{1}{21}$

$n = 4$

$\dfrac{1}{12} - \dfrac{1}{15} - \dfrac{1}{21} + \dfrac{1}{24}$

Er blijven enkel 4 termen over:

$n = 1$

$\dfrac{1}{3} - \cancel{\dfrac{1}{6}} - \dfrac{1}{12} + \cancel{\dfrac{1}{15}}$

$n = 2$

$\cancel{\dfrac{1}{6}} - \cancel{\dfrac{1}{9}} - \cancel{\dfrac{1}{15}} + \cancel{\dfrac{1}{18}}$

$n = 3$

$$\cancel{\frac{1}{9}} - \cancel{\frac{1}{12}} - \cancel{\frac{1}{18}} + \cancel{\frac{1}{21}}$$

$n = 4$

$$\cancel{\frac{1}{12}} - \frac{1}{15} - \cancel{\frac{1}{21}} + \frac{1}{24}$$

$$= \frac{1}{3} - \frac{1}{12} - \frac{1}{3(n+1)} + \frac{1}{3(n+4)}$$

$$= \frac{1}{4} - \frac{1}{3(n+1)} + \frac{1}{3(n+4)}$$

$$= \frac{3(n+1)(n+4) - 4(n+4) + 4(n+1)}{12(n+1)(n+4)}$$

$$= \frac{3(n^4 + 5n + 4) - 4n - 16 + 4n + 4}{12(n^2 + 5n + 4)}$$

$$= \frac{3n^4 + 15n \cancel{+ 12} \cancel{- 4n} \cancel{- 16} \cancel{+ 4n} \cancel{+ 4}}{12(n^2 + 5n + 4)}$$

$$= \frac{\cancel{3}(n^2 + 5n)}{\cancel{12}(n^2 + 5n + 4)}$$

$$= \frac{n^2 + 5n}{4n^2 + 20n + 16}$$

References